LAB 1: Problem Statement: Public bus live tracking & crowding estimator

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SECTION: G

Problem Number:24

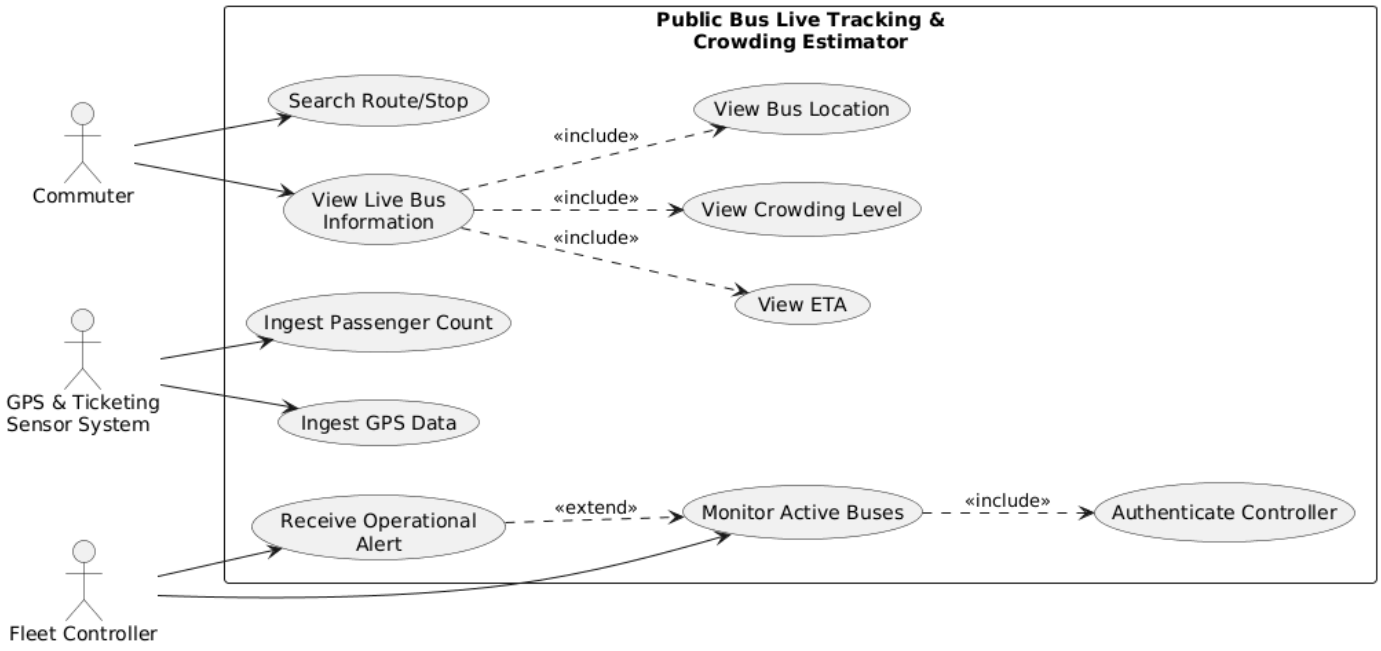
FUNCTIONAL REQUIREMENTS TABLE:

ID & Priority	Requirement Description	Acceptance Criteria	Rationale
FR-001 — High	The system shall ingest GPS coordinates from active buses at least every **5 seconds** and update their current location.	**Pass:** A bus location is updated using the latest valid GPS coordinate within 5 seconds. **Fail:** The displayed location is older than 5 seconds when valid GPS data is available.	Provides accurate real-time bus tracking and improves commuter confidence in the service.
FR-002 — High	The system shall calculate and display the **estimated time of arrival (ETA)** of a tracked bus at its upcoming route stops.	**Pass:** ETA is recalculated using the bus's current location, distance, and estimated travel speed. **Fail:** ETA does not update after a change in bus location.	Helps commuters plan their journeys and reduces uncertainty about bus arrival times.
FR-003 — High	The system shall estimate and display the **crowding level** of each tracked bus using passenger-count data received from ticketing sensors.	**Pass:** The system classifies crowding as **Low, Moderate, or High** according to configured passenger thresholds. **Fail:** No crowding level is displayed when valid passenger data exists.	Helps commuters choose less crowded buses and enables better monitoring of passenger capacity.
FR-004 — High	The system shall allow commuters to search for a **bus route or stop** and view the live location, ETA, and crowding level of relevant buses.	**Pass:** A valid route/stop search displays active buses with their latest location, ETA, and crowding level. **Fail:** Relevant active buses are not displayed.	Provides commuters with the key information needed to make informed travel decisions.
FR-005 — Medium	The system shall allow fleet controllers to **monitor active buses** and receive alerts when a bus has missing telemetry or an abnormal crowding level.	**Pass:** An alert is generated when telemetry exceeds the configured timeout or crowding exceeds the configured threshold. **Fail:** No alert is generated for a detected abnormal condition.	Helps fleet controllers identify operational issues and overcrowding quickly, improving fleet management.

Non-FUNCTIONAL REQUIREMENTS TABLE:

ID & Priority	Requirement Description	Acceptance Criteria	Rationale
NFR-001 — High	The tracking engine shall process telemetry from up to **1,000 active buses simultaneously** and update bus locations and ETAs within **3 seconds** of receiving valid telemetry.	**Pass:** Load testing with 1,000 simulated buses maintains location/ETA update latency of **≤3 seconds for at least 95% of updates**. **Fail:** More than 5% of updates exceed 3 seconds.	Ensures the platform remains responsive and reliable during peak transit activity and future city-wide expansion.
NFR-002 — High	The system shall authenticate fleet-controller access, protect transit and passenger-related data using secure communication, and maintain at least **99.5% monthly availability**.	**Pass:** Unauthorized users cannot access controller functions, data is transmitted securely, and monthly availability is **≥99.5%**. **Fail:** Unauthorized access is permitted or availability falls below 99.5%.	Protects operational data from unauthorized access while ensuring the service remains dependable for commuters and transit authorities.

Use-Case Diagram: Public bus live tracking & crowding estimator



Use Case Flow Specification:

Use Case ID - UC-001

Use Case Name - View Live Bus Information

Primary Actor – Commuter, Fleet Controller

Secondary Actor - Bus GPS & Ticketing Sensors

Goal - The commuter wants to view the current location, estimated arrival time, and crowding level of a bus.

Preconditions

1. The commuter has access to the Public Bus Live Tracking & Crowding Estimator.
2. The selected bus route or stop exists in the system.
3. The system is receiving valid bus telemetry and passenger-count data.

Postconditions

1. The commuter is shown the latest available bus location.
2. The commuter is shown the estimated arrival time at the selected stop.
3. The commuter is shown the current crowding level of the bus.

Main Success Scenario

1. The commuter opens the bus tracking system.

2. The system displays the route/stop search interface.

3. The commuter searches for a bus route or stop.

4. The system identifies active buses serving the selected route/stop.

5. The system retrieves the latest GPS location of the selected bus.

6. The system calculates the bus's estimated arrival time using its current location, route distance, and estimated travel speed.

7. The system retrieves the latest passenger-count information from ticketing sensors.

8. The system determines the bus crowding level as Low, Moderate, or High.

9. The system displays the bus's current location, ETA, and crowding level to the commuter.

10. The commuter views the live bus information.

Alternate Flow

AF-01: GPS/Telemetry Data Unavailable

At Step 5, if the system does not receive valid GPS data within the configured timeout:

1. The system detects that the bus telemetry is unavailable.
2. The system displays the **last known bus location** and indicates that live tracking is temporarily unavailable.
3. The system cannot provide a reliable updated ETA.
4. The system displays an appropriate status message to the commuter.
5. When new valid GPS data is received, the system resumes live tracking and recalculates the ETA.